

NLP Lab Exp 5

September 2026

1 Part-of-Speech (POS) Tagging

Part-of-Speech (POS) tagging is the process of assigning a grammatical category to each word in a sentence based on its context.

For example:

The small boy quickly ate an apple.

The POS tags are:

Word	POS Tag	Grammatical Role
The	DT	Determiner
small	JJ	Adjective
boy	NN	Noun
quickly	RB	Adverb
ate	VBD	Verb
an	DT	Determiner
apple	NN	Noun

Thus, POS tagging answers the question:

What is the grammatical role of each word?

1.1 Why is Context Important?

The same word can have different grammatical roles depending on its context.

For example:

I read a **book**.

Here, **book** is a noun.

Please **book** a ticket.

Here, **book** is a verb.

Therefore, POS tagging does not simply look up a word in a dictionary. It considers the context in which the word appears.

2 Standard Text vs. Social Media Text

POS tagging is generally easier for standard, well-formed text because it follows conventional grammar, spelling, and punctuation.

2.1 Standard Text

Example:

I am learning Natural Language Processing.

Standard text generally has:

- Correct spelling
- Standard grammar
- Proper punctuation
- Well-formed sentences
- Clearly separated words

Examples of standard text include:

- News articles
- Books
- Research papers
- The Wall Street Journal corpus
- Penn Treebank

2.2 Twitter / Social Media Text

Social media text is often informal and noisy.

For example:

Going 2 the market rn #shopping @John

It may contain:

- Misspellings
- Abbreviations
- Emojis
- Hashtags

- User mentions
- URLs
- Slang
- Informal grammar
- Special symbols

For example:

- 2 → to
- rn → right now
- smh → shaking my head
- #NLP → hashtag
- @John → user mention

Because of these characteristics, normal tokenizers may incorrectly split social-media expressions.

Specialized tokenizers such as NLTK's `TweetTokenizer` are therefore useful. For example:

@John I'm learning #NLP

can ideally be tokenized as:

@John | I'm | learning | #NLP

3 Tagsets

A **tagset** is a collection of labels used to represent the grammatical categories of words in a sentence.

There are two commonly used POS tagsets:

3.1 Penn Treebank Tagset

The Penn Treebank tagset uses **fine-grained tags**. It provides detailed information about the grammatical role and form of a word.

Some common tags are:

Tag	Meaning	Example
NN	Singular noun	book
NNS	Plural noun	books
VB	Base form of verb	eat
VBD	Past-tense verb	ate
VBZ	3rd-person singular present verb	eats
JJ	Adjective	beautiful
RB	Adverb	quickly
DT	Determiner	the

For example:

Dogs run.

The POS tags are:

Dogs/NNS run/VB

Here, NNS indicates that **Dogs** is a plural noun, while VB indicates that **run** is the base form of a verb.

3.2 Universal POS Tagset

The Universal POS tagset uses **coarse-grained and standardized categories**.

Some common categories are:

Tag	Meaning	Example
NOUN	Noun	book
VERB	Verb	run
ADJ	Adjective	beautiful
ADV	Adverb	quickly
PRON	Pronoun	he
DET	Determiner	the
ADP	Adposition	in
PUNCT	Punctuation	.

For example:

Dogs run quickly.

The Universal POS tags are:

Dogs -> NOUN
run -> VERB
quickly -> ADV

Unlike the Penn Treebank tagset, Universal POS does not distinguish between singular and plural nouns or between different forms of verbs.

3.3 Penn Treebank vs. Universal POS

Feature	Penn Treebank	Universal POS
Type	Fine-grained	Coarse-grained
Level of detail	High	Lower
Noun example	NN, NNS	NOUN
Verb example	VB, VBD, VBZ	VERB
Main advantage	Detailed information	Simple and standardized

Easy way to remember:

- **Penn Treebank:** More details about the word.
- **Universal POS:** General grammatical category of the word.

4 Chunking (Shallow Parsing)

4.1 What is Chunking?

Chunking is the process of grouping words in a sentence into syntactically related phrases.

POS tagging tells us:

What is each word?

Chunking tells us:

Which words belong together?

Consider:

The small boy ate an apple.

After chunking, the sentence can be divided into meaningful phrases:

[The small boy] [ate] [an apple]

Here:

- [The small boy] is a Noun Phrase (NP).
- [ate] is a verb.
- [an apple] is a Noun Phrase (NP).

4.2 Why is Chunking Called Shallow Parsing?

Chunking is also called **Shallow Parsing** because it identifies basic phrases without building a complete syntactic structure.

For example:

[The small boy] [ate] [an apple]

A full syntactic parser would provide a hierarchical structure such as:

Sentence
Noun Phrase
The small boy

Verb Phrase
ate
Noun Phrase
an apple

Therefore:

Chunking identifies important phrases, while full parsing determines the complete grammatical structure of the sentence.

5 Noun Phrase (NP) Chunk

A **Noun Phrase (NP)** is a group of words centered around a noun.

The noun is the main word, or **head**, of the phrase. Other words may specify or describe the noun.

5.1 Examples

the boy
the small boy
a beautiful house
the very small boy
an expensive new car

In each example, there is a main noun:

Noun Phrase	Head Noun
the boy	boy
the small boy	boy
a beautiful house	house
the very small boy	boy
an expensive new car	car

5.2 Basic NP Pattern

A simple Noun Phrase can be represented as:

Determiner + Adjective(s) + Noun

This pattern means that a Noun Phrase may contain:

- **Determiner:** Specifies or identifies the noun, such as **the**, **a**, or **an**.
- **Adjective(s):** Describes the noun. There may be zero, one, or more adjectives.
- **Noun:** The main or head word of the Noun Phrase.

For example:

the + small + boy

Here:

the -> Determiner
small -> Adjective
boy -> Noun

Therefore:

[**The small boy**]

is a **Noun Phrase (NP)**.

5.3 Components of a Noun Phrase

5.3.1 Determiner

A **determiner** specifies or identifies the noun.

Common determiners include:

the
a
an
this
that
these
those

Examples:

the boy
a car
an apple
this book
those students

The determiner is often optional.

5.3.2 Adjective

An **adjective** describes or provides additional information about a noun.

Examples:

small boy
beautiful house
expensive car
difficult problem

Multiple adjectives can occur together:

a beautiful large house
an expensive new car
the small intelligent student

5.3.3 Noun

The **noun** is the central element of a Noun Phrase.

Examples:

boy
house
car
student
problem

Consider:

[the small boy]

Here:

the -> Determiner
small -> Adjective
boy -> Noun

The word boy is the head noun of the phrase.

5.4 Noun Phrase Examples

5.4.1 Example 1

The boy is playing.

Noun Phrase:

[The boy]

Structure:

Determiner + Noun

5.4.2 Example 2

The small boy is playing.

Noun Phrase:

[The small boy]

Structure:

Determiner + Adjective + Noun

5.4.3 Example 3

The very small boy is playing.

Noun Phrase:

[The very small boy]

Here:

the -> Determiner
very -> Adverb modifying the adjective
small -> Adjective
boy -> Noun

The structure is:

Determiner + Adverb + Adjective + Noun

5.4.4 Example 4

She bought an expensive new car.

Noun Phrase:

[an expensive new car]

Structure:

Determiner + Adjective + Adjective + Noun

Here:

an -> Determiner
expensive -> Adjective
new -> Adjective
car -> Noun

5.5 Finding Noun Phrases Using POS Tags

Noun phrases can be identified using POS tags.

Consider:

The small boy ate an apple.

The POS-tagged sentence is:

The/DT small/JJ boy/NN ate/VBD an/DT apple/NN

We can identify the noun phrases as:

[The/DT small/JJ boy/NN]

[an/DT apple/NN]

Therefore:

[The small boy] [ate] [an apple]

The two noun phrases are:

[The small boy]

[an apple]

5.6 NP Chunking Using a Regular Expression

In NLP, a chunking grammar can be defined using a **regular expression over POS tags**.

For example:

NP: {<DT>?<JJ>*<NN.*>+}

This rule can be understood as follows:

Pattern	Meaning
<DT>?	Optional determiner
<JJ>*	Zero or more adjectives
<NN.*>+	One or more noun-related tags

Therefore, the rule can identify phrases such as:

the boy

the small boy

a beautiful house

an expensive new car

For example, for:

The/DT small/JJ boy/NN

the pattern matches:

DT + JJ + NN

and produces:

[The small boy]

5.7 NP Chunk vs. Individual Noun

It is important to distinguish between a **noun** and a **noun phrase**.

Consider:

The small boy is running.

The word:

boy

is a **noun**.

The complete group:

[The small boy]

is a **Noun Phrase**.

Therefore:

Noun -> boy

Noun Phrase -> The small boy

5.8 Key Points

- A **Noun Phrase (NP)** is a group of words centered around a noun.
- The noun is called the **head noun**.
- An NP may contain a determiner, adjectives, and a noun.
- Determiners and adjectives are not always required.
- A simple NP pattern is:

Determiner + Adjective(s) + Noun

- POS tags can be used to identify Noun Phrases.
- In chunking, an NP is represented as a group of related words.

A Noun Phrase is a group of words that works together as a noun in a sentence.

Examples:

[the boy]

[the small boy]

[a beautiful house]

[an expensive new car]

6 Verb Phrase (VP) Chunk

A **Verb Phrase (VP)** is a group of words centered around a verb.

The verb is the main word, or **head**, of the phrase. Other words may provide additional information about the action or state.

6.1 Examples

quickly ate
was running
has completed

6.2 Simple VP Pattern

A simple Verb Phrase pattern may contain:

Adverb + Verb

For example:

quickly + ate

Therefore:

[quickly ate]

can be identified as a Verb Phrase (VP) or a verb-related chunk, depending on the chunking grammar being used.

Other examples include:

[quickly ate]
[was running]
[has completed]

Here:

- quickly modifies the verb **ate**.
- **was running** contains an auxiliary verb **was** and a main verb **running**.
- **has completed** contains an auxiliary verb **has** and a main verb **completed**.

The exact structure of a Verb Phrase chunk depends on the grammar rules defined for the chunking parser.

6.3 NP and VP Chunks

Consider:

The small boy quickly ate.

The chunks can be represented as:

[The small boy] [quickly ate]
NP VP

Thus:

- **NP chunk** identifies a group centered around a noun.
- **VP chunk** identifies a group centered around a verb.

7 Grammar Specifications for Chunking

Chunking grammars are commonly defined using **regular expressions over POS tags**.

For example, an NP grammar might specify:

NP: {<DT>?<JJ>*<NN.*>+}

This can be interpreted as:

- <DT>? → optional determiner
- <JJ>* → zero or more adjectives
- <NN.*>+ → one or more noun-related tags

For example:

The/DT small/JJ boy/NN

matches:

DT + JJ + NN

and produces:

[The small boy]

The exact regular expression depends on the POS tagset and the type of phrase that we want to identify.

8 Chinking

Chinking is the process of removing unwanted parts from a chunk.

In simple terms:

- **Chunking:** What should be included?
- **Chinking:** What should be excluded?

Suppose we have:

The small boy in the classroom

We may want to identify:

[The small boy]

while excluding:

in the classroom

The excluded portion is called a **chink**.

Therefore:

Chunking builds a phrase, while chinking removes unwanted elements from a potential phrase.

9 IOB Tagging Scheme

IOB tagging is a common method for representing chunks.

IOB stands for:

- **B = Beginning**
- **I = Inside**
- **O = Outside**

Consider:

The small boy ate an apple.

Suppose the noun phrases are:

[The small boy] ate [an apple]

The IOB representation is:

Word	IOB Tag
The	B-NP
small	I-NP
boy	I-NP
ate	O
an	B-NP
apple	I-NP

9.1 Meaning of the Tags

9.1.1 B – Beginning

Indicates the first word of a chunk.

For example:

The/B-NP

9.1.2 I – Inside

Indicates a word that occurs inside an existing chunk.

For example:

small/I-NP

boy/I-NP

9.1.3 O – Outside

Indicates that the word does not belong to the chunk.

For example:

ate/O

10 Syntactic Parsing (Full Parsing)

Full syntactic parsing goes beyond POS tagging and chunking.

It analyzes a sentence to discover its **complete grammatical structure**.

For example:

The small boy ate an apple.

A full parser tries to determine:

- Which words form phrases?
- Which phrase is the subject?
- Which phrase is the object?
- What is the main verb?
- How are different parts of the sentence related?
- What is the hierarchical structure of the sentence?

Therefore, full parsing answers the question:

How are all the words and phrases grammatically related?

There are two major approaches:

1. Constituency Parsing
2. Dependency Parsing

11 Constituency Parsing

Constituency parsing is also called **phrase-structure parsing**.

It divides a sentence into **nested grammatical constituents or phrases**.

Consider:

The small boy ate an apple.

A simplified structure is:

Sentence

Noun Phrase (NP)

The

small boy

Verb Phrase (VP)

ate

Noun Phrase (NP)

an

apple

The important idea is **hierarchy**.

The sentence contains:

Sentence

which contains:

NP + VP

The VP contains:

Verb + NP

Thus, constituency parsing represents the sentence as a **nested tree of phrases**.

12 Dependency Parsing

Dependency parsing takes a different approach.

Instead of focusing primarily on phrases, it focuses on **relationships between individual words**.

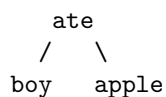
Consider:

The boy ate an apple.

The main word is:

ate

Other words are connected to it through grammatical relationships.
A simplified dependency structure is:



The relationships can be represented as:

```
boy    -> ate
apple  -> ate
```

Common dependency labels include:

- `nsbj` → nominal subject
- `obj` → object
- `amod` → adjectival modifier
- `det` → determiner
- `advmod` → adverbial modifier

For example:

The smart boy ate an apple quickly.

A simplified dependency representation is:

```
The      -> boy      (det)
smart    -> boy      (amod)
boy       -> ate      (nsbj)
apple    -> ate      (obj)
an        -> apple    (det)
quickly  -> ate      (advmod)
```

The main verb **ate** acts as the central word, and other words are connected to it directly or indirectly.